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Ph.D. Candidate

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OBJECTIVES

Short-term: To pursue a postdoctoral position that broadens my expertise in gene-editing and computational biology, while building the scientific independence, leadership skills, and collaborative network necessary to launch a successful academic career.

Long-term: To establish an independent research laboratory that integrates experimental and computational innovations to decipher the biological mechanisms underlying gene editing, and to develop clinically translatable platforms that realize precise, safe, and durable therapeutic editing.

EDUCATION

Ph.D. in **Bioengineering, Rice University,** *Houston, TX* *Aug 2020 – Aug 2026 (expected)*
M.Sc. in **Bioinformatics, Georgia Institute of Technology,** *Atlanta, GA* *Aug 2018 – Dec 2019*
B.Sc. in **Bioengineering, Beijing Institute of Technology,** *Beijing, China* *Sep 2013 – Jun 2017*

GRADUATE RESEARCH

Bao Lab, Rice University *Sep 2020 – Present*
Advisor: Gang Bao, Ph.D. | Professor, Department of Bioengineering, Rice University
Integrating experimental and computational methods to decode gene editing outcomes in human disease models.

➢ Fine-Mapping Inflammatory Bowel Disease (IBD) eQTL Regulatory Variants via eCROP-seq and Base Editing [Pub. 1]

- Developed a moderate-throughput ***eCROP-seq*** (expression CRISPR droplet sequencing) workflow using a lentiviral delivery system, optimizing scalability for high-content eQTL functional screens across diverse immune cell contexts — HL60, Jurkat, Macrophage, Neutrophil, and primary T cells.
- Screened over 4,000 common variant candidates from IBD GWAS credible sets overlapping ATAC-seq peaks, including 110 negative controls outside eQTL signal, and validated over 400 causal variants linked to gene expression changes.
- Screened rare variants (MAF < 0.01) across 89 IBD-associated genes; identified one-third rare variants that significantly alter target gene expression, with multiple significant hits per gene.
- Performed HL60 macrophage and neutrophil differentiation to examine cell-type-specific regulatory effects of validated eQTL causal variants, confirming the cell-tissue specificity in eQTL variants.

- Applied adenine base editing (ABE) and cytosine base editing (CBE) screening to introduce precise single-nucleotide changes at multiple candidate and control sites, directly linking genotype to transcriptional output at single-cell resolution.

➢ **Long-Read Sequencing-Based Decoding of Large Gene Modifications Induced by CRISPR/Cas9 [Pubs. 2, 3, 4, 7, 8]**

- Designed and implemented ***LV_caller***, a bioinformatics pipeline for systematic detection and quantification of large unintended gene modifications (deletions, insertions, inversions) from CRISPR/Cas9 editing using Single Molecule, Real-Time Sequencing (SMRT-seq).
- Improved the computational analysis procedure of ***LongAmp-Seq***, an in-house NGS-based approach, achieving comparable sensitivity to SMRT-seq for profiling large unintended gene modifications.
- Demonstrated that CRISPR/Cas9 induces large deletions at high frequencies (11.7–35.4%) at therapeutically critical loci (HBB, HBG, BCL11A, PD-1) in HSPCs and T cells, with direct implications for gene therapy safety assessment. [**Pub. 2**]
- Applied ***LV_caller*** to profile large deletions in a Sickle-HUDEP2 GFP/BFP reporter cell model [Pub. 5] and in SaCas9 vs. SpCas9 editing comparisons [Pub. 11], contributing to multiple long-read sequencing studies of CRISPR/Cas9 editing outcomes [**Pubs. 4, 7**].

➢ **scGoT-EDIT: A Single-Cell Genotyping and RNA Sequencing Platform** [In preparation]

- Developed a single-cell mRNA genotyping and analysis platform, named Single-Cell Genotyping of Transcriptome for EDITing outcomes (***scGoT-EDIT***), integrating CRISPR editing outcomes with 10x Genomics transcriptomic readouts, enabling simultaneous genotype-to-transcriptome resolution at the single-cell level.
- Conducted comprehensive GoT analysis of CRISPR/Cas9 editing outcomes across erythroid differentiation stages in SCD patient hematopoietic stem and progenitor cells (HSPCs).
- Linked diverse base editing outcomes to transcriptomic analysis in primary T cells.

➢ **Machine Learning Models for CRISPR/Cas Off-Target Prediction**

- Benchmarked state-of-the-art deep learning models for predicting off-target cleavage sites of Cas9 nucleases and published a relating technical review. [**Pub. 3**]
- Evaluated transfer learning strategies to improve cross-dataset generalizability, identifying key factors limiting model performance on unseen guide sequences. [**Pub. 8**]

➢ **Other Selected Contributions**

- Developed ***GISA-seq*** (Genome-wide Integration Site Analysis by Sequencing), an anchored PCR-based NGS assay with UMI-tagged adpators for unbiased genome-wide retrieval of AAV-donor integration sites; applied to demonstrate selective on-target HDR enrichment by the Repair Drive platform in mouse liver. [**Pub. 6**]
- Performed single-cell RNA-seq analysis on iPSC-derived pancreatic islets carrying CFTR mutations, characterizing impaired glucagon secretion and identifying cell-type-specific transcriptional dysregulation linked to cystic fibrosis-related diabetes (CFRD). [In preparation]
- Performed bioinformatics analysis, especially iGUIDE-seq for CRISPR/Cas9 off-target, for Baylor/Rice Genome Editing Testing Center (GETC).
- Provided ongoing bioinformatics consultation to the Genetic Design and Engineering Center (GDEC) at Rice University.
- Served as a research member at Baylor/Rice Genome Editing Testing Center (GETC)

PUBLICATIONS

Selected Highlight

1. Iannetta E†, **Cao M†**, Lee C, Liu A, Park SH, Dechanun S, Betancourth D, Brown M, Serrano-Marin L, Bao G, Gibson G. Haplotype-embedded regulatory variation shapes gene expression and immune associations. *Science*. Under revision (2026). *†Co-first authors, contributed equally | Under revision at **Science** |* (BioRxiv doi: https://doi.org/10.1101/2025.01.30.635675)
2. Park SH, **Cao M**, Pan Y, Davis TH, Deshmukh H, Fu Y, Treangen T, Sheehan VA, Bao G. Comprehensive analysis of unintended large gene modifications induced by CRISPR/Cas9 gene editing. *Sci Adv*. 2022;8(42):eabo7676. DOI: 10.1126/sciadv.abo7676.
3. **Cao M**, Brennan A, Lee CM, Park SH, Bao G. Deep learning-based models for CRISPR/Cas off-target prediction. *Small Methods*. 2025;9(7):e2500122. DOI: 10.1002/smtd.2500122.

Published

4. Chuecos MA, Park SH, Bhakta MM, Too-Chiobi U, Betancourth D, **Cao M**, De Giorgi M, Walkey CJ, Tiwari A, Godin B, Assini JM, Palmer DJ, Ng P, Boffa MB, Koschinsky ML, Bao G, Lagor WR. Cytosine base editing of *LPA* in transgenic mice averts large deletions. *Mol Ther*. In press (2026). DOI: 10.1016/j.ymthe.2026.02.049.
5. Karsenty CL, Betancourth D, **Cao M**, Pham QK, Park SH, Bao G. Complex HBB gene editing outcomes revealed by a fluorescent reporter cell model. *Mol Ther Nucleic Acids*. 2026. DOI: 10.1016/j.omtn.2026.102854.
6. De Giorgi M, Park SH, Hurley A, Saxena L, **Cao M**, Chuecos MA, Walkey C, Doerfler A, Furgurson MN, Hyde S, Chickering T, Lefebvre S, Qin J, Bissig K, Bao G, Castoreno A, Jadhav V, Lagor WR. In vivo expansion of gene-targeted hepatocytes enables safe and durable transgene expression. *Sci Transl Med*. 2025;17:eadk3920. DOI: 10.1126/scitranslmed.adk3920.
7. Park SH, **Cao M**, Bao G. Detection and quantification of large unintended on-target gene modifications due to CRISPR/Cas9 editing. *Curr Opin Biomed Eng*. 2023;51(5):e26. DOI: 10.1016/j.cobme.2023.100478.
8. Kota PK, Pan Y, Vu HA, **Cao M**, Baraniuk RG, Bao G. The need for transfer learning in CRISPR-Cas off-target scoring. *bioRxiv*. Preprint. 2021. DOI: 10.1101/2021.08.28.457846.
9. Liu H, **Cao M**, Wang Y, Lv B, Li C. Bioengineering oligomerization and monomerization of enzymes: learning from natural evolution to matching the demands for industrial applications. *Crit Rev Biotechnol*. 2019. DOI: 10.1080/07388551.2019.1711014.
10. Zhang Z, **Cao M**, Li J, Li C, Liu H. Strategies and application of highly efficient secretion of proteins by microorganisms. *Chem Ind Eng Prog* [Chinese]. 2017. DOI: 10.16085/j.issn.1000-6613.2017-2100.

Under Revision

11. Yoo BY, **Cao M**, Karsenty C, Betancourth D, Pham QK, Zhang Y, Pendergast MA, Sheehan VA, Park SH, Bao G. SaCas9 and AsCas12a have highly efficient and specific gene editing in hematopoietic stem and progenitor cells. Under revision at *Cell Biomaterials*. (2026).

PROFESSIONAL SKILLS HIGHLIGHTS

➢ **Wetlab**

- Gene-editing: CRISPR/Cas9, adenine/cytosine base editors, lentiviral production and delivery
- Sequencing: NGS library prep, bulk RNA-seq, 10x Genomics scRNA-seq, SMRT-seq.
- Molecular and synthetic biology: Genetic circuit design and plasmid construction, flow cytometry, ddPCR.
- Biochemical: ELISA, Western blot, SDS/Native-PAGE, HPLC, protein purification, ultracentrifugation.

➢ **Bioinformatics Analysis**

- Single-cell RNA sequencing: Seurat, CellRanger, scVelo, CellRank, GoT, trajectory/differential expression analysis

- Genomics: CRISPR off-target analysis (GUIDE-seq, CAST-seq), integration site analysis (iGUIDE), structural variant detection (LV_caller, SMRT-seq), whole-genome sequencing (WGS).
- Bulk RNA sequencing: Salmon, DESeq2
- Machine learning/deep learning models: CNN, RNN, Attention and hybrid models (Tensorflow).
- Programming: Python, R, SQL, Bash/Linux; statistical modeling; dynamic modeling.

➢ **Languages**: Chinese (Mandarin, native), English (professional fluency).

AWARDS & HONORS

➢ **Rice University**

- 2026 BIOE Excellence in Research Award.
  May 2026 Recognizes sustained research excellence across a student's doctoral tenure in the Rice Bioengineering PhD program.

➢ **Georgia Institute of Technology**

- Graduate Research Assistantships (GRA) by Computational Biology Faculty Steering Committee, School of Biological Sciences
  Jun 2018

➢ **Beijing Institute of Technology**

- The Gold Award, International Genetically Engineered Machine Competition *Nov 2014 & Sep 2015 & Nov 2018*
- Nominated for Best New Application Project in iGEM
  Sep 2015
- The First Award & the Special Award, "Century Cup" Competition of BIT *May 2015 & May 2016*
- The Second Award, "Challenge Cup" Beijing College Students Scientific Works Competition *Jul 2015*

CONFERENCE PRESENTATIONS

- **Cao M**, Park SH, Bao G. Single-cell genotyping and transcriptomic analysis of CRISPR/Cas9-edited hematopoietic stem and progenitor cells for treating sickle cell disease. Poster presentation. *28th Annual Meeting of the American Society of Gene and Cell Therapy (ASGCT)*; May 13–17, 2025; New Orleans, LA.
- **Cao M**, Park SH, Bao G. High-throughput genotype and functional analysis of CRISPR/Cas9-edited hematopoietic stem and progenitor cells using single-cell RNA-seq. Poster presentation. *BMES Cellular and Molecular Bioengineering (CMBE) Conference*; January 2–6, 2024; San Juan, Puerto Rico.
- **Cao M**, Park SH, Bao G. High-throughput genotype and functional analysis of CRISPR/Cas9-edited hematopoietic stem and progenitor cells using single-cell genotyping to transcriptome (GoT). Poster presentation. *26th Annual Meeting of the American Society of Gene and Cell Therapy (ASGCT)*; May 16–20, 2023; Los Angeles, CA.
- **Cao M**, Park SH, Bao G. A pipeline for improving off-target identification and validation after GUIDE-seq. Poster presentation. *Biomedical Engineering Society (BMES) Annual Meeting*; October 6–9, 2021; Orlando, FL.

MASTER & UNDERGRADUATE RESEARCH

Lu Lab, Rice University (Summer Research) *Jun 2020 – Sep 2020*
Advisor: George Lu, Ph.D. | Assistant Professor, Department of Bioengineering

- Mined prokaryotic genomes to identify gas vesicle protein (Gvp) clusters; analyzed N/C-terminal extensions in GvpA proteins computationally; cloned and purified Gvp clusters for functional characterization.

Lieberman Lab, Georgia Institute of Technology *Oct 2018 – Mar 2020*

Advisor: Raquel L. Lieberman, Ph.D. | Professor, School of Chemistry and Biochemistry

➢ **Phylogenetic and Evolutionary Analysis of the Olfactomedin Protein Domain**

- Mined NCBI, UniProt, eggNOG, and JGI databases via BLAST/HMMER; refined dataset using CD-HIT and VSearch; built MSAs (MUSCLE, ClustalO, Promals3D) and phylogenetic trees (MEGA, ggtree) with ML, Bayesian, NJ, and ME methods.
- Annotated domain architectures (CD-Search, InterProScan, SignalP, TMHMM) and identified conserved metal-binding motifs (MEME, IonCom); analyzed gene divergence (BEAST), selection pressure (PAML), and protein structures (Phyre2, PyMOL); automated workflows in Python and R.

Li Lab, Beijing Institute of Technology (BIT) [Pub. 9, 10] *Feb 2017 – Jun 2018*

Advisor: Chun Li, Ph.D. | Professor, School of Chemistry and Chemical Engineering

➢ **Enzyme Depolymerization Engineering (Undergraduate Thesis)**

- Thesis: *Analysis of Depolymerization Structure and Catalytic Properties of Penicillium purpurogenum β-glucuronidase (PGUS)* (with postdoc Hu Liu).
- Designed mutation strategies based on structural analysis and molecular docking to depolymerize PGUS from its native homo-tetramer to catalytically active dimers and monomers.
- Successfully engineered at least 4 active PGUS dimer variants and active monomer inclusion bodies, improving per-subunit catalytic efficiency.

➢ **International Genetically Engineered Machine (iGEM) Competition**

- 2015 BIT-China — *pH Balance Machine*, Gold Award: Designed alkali-resistance gene circuits; conducted high-throughput error-prone PCR mutagenesis to build a promoter mutant library; validated integrated strains with mathematical modeling. Served as Human Practices team leader at the Asian iGEM Conference, Taiwan.
- 2014 BIT-China — *E.co-Lock*, Gold Award: Designed sRNA-based negative feedback regulation circuits; contributed to computational modeling; presented the project as a speaker at iGEM 2014 in Boston.

TEACHING & MENTORING EXPERIENCE

Rice University — Bao Lab *Sep 2020 – Present*

- Mentored at least 5 undergraduate researchers and several technicians in CRISPR gene editing experiments, single-cell sequencing workflows, and bioinformatics analysis.
- Worked as teaching assistant in three classes including Gene Therapy (BIOE 422/522), Bioengineering Fundamentals (BIOE 252), Lab Module in Tissue Culture (BIOE 342), leading project discussions and giving lectures.

Li Lab, BIT — Short-term Lecturer *Jan 2018 – Feb 2018*

- Tutored high school students in synthetic biology fundamentals and core molecular biology techniques (PCR, bacterial transformation, protein expression).

iGEM Team Advisor — BIT-China 2018 (Gold Award) *Mar 2018 – Oct 2018*

- Served as project designer, competition director, experiment trainer, and academic consultant for the project "An antioxidant-testing device in yeast."

iGEM Team Advisor — CIEI-BJ High School (Silver Award) *Mar 2018 – Oct 2018*

International Teenager Competition and Communication Center (ITCCC)

- Advised a high school team on project design and experimental execution for "Using oxidase to degrade aflatoxin."